

Anuj Shukla

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EDUCATION

MIT Manipal — B.Tech, Computer and Communication Engineering

Aug 2023 – Present

SKILLS & TECHNOLOGIES

AI / ML: RAG Pipelines, AI Agents, LangChain, LlamaIndex, Vector Databases (FAISS, ChromaDB), Embeddings, Prompt Engineering, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy

Languages: Python, JavaScript, TypeScript, Java, C

Frameworks / Libraries: FastAPI, React.js, Node.js, Express.js, Tailwind CSS

Databases: MySQL, PostgreSQL, MongoDB, Supabase, SQLite

Tools / Platforms: Git, GitHub, VSCode, Netlify, Google Colab, FastAPI, LLM APIs (Gemini)

PROJECTS

CMF2-Rec: Cascaded Meta-Feature Fusion Recommender | Python, PyTorch, Scikit-learn, FastAPI, OpenML

- Developed a cascaded meta-learning framework recommending optimised ML pipelines (preprocessing + model + hyperparameters) using a 52-dimensional statistical meta-feature vector; leveraged **structured LLM prompting techniques** (Chain-of-Thought, Tree-of-Thought, role and context framing) during system design and feature engineering.
- Engineered a dual-layer architecture combining a deterministic rule-based compatibility filter with a **Schema Graph Convolutional Network (GCN)** for deep inter-column dependency mapping.
- Achieved **83.3% Top-1 and 100% Top-3 accuracy** on held-out OpenML benchmarks; exposed inference via a FastAPI REST API with **LIME + SCM counterfactual justifications** for every recommendation.

MinTrack: Full-Stack Academic Productivity System | React 19, Supabase, TypeScript, Tailwind CSS, Web Workers

- Architected an **offline-first sync engine** using Supabase and LocalStorage, enabling seamless cross-device persistence with automatic conflict resolution between cached data and a cloud-hosted PostgreSQL database.
- Engineered a high-precision Pomodoro timer via **Web Workers** for sub-millisecond accuracy when backgrounded, paired with the Web Audio API for non-blocking auditory feedback; built a debounced CRUD sync queue with retry logic for eventual consistency under network instability.
- Implemented a dynamic progress engine that auto-recalculates daily study targets and carryover pressure across multi-day absences using timestamped logic.

FoodKart: Decision-Support Android Food Discovery System | Java, Android SDK, SQLite, AHP, Fuzzy TOPSIS | *Manuscript under review*

- Engineered a native Android prototype ranking restaurants across competing factors (price, quality, distance) using a **multi-criteria recommendation pipeline**.
- Implemented a hybrid mathematical model combining **AHP** for preference weight derivation and **Fuzzy TOPSIS** for robust ranking under uncertainty.
- Built a fully offline architecture using SQLite for local persistence and Android Architecture Components for real-time ranking updates.

E2E Encrypted Chat Application | React, TypeScript, Web Crypto API, Netlify | [Live Demo](#)

- Built a real-time E2E encrypted chat app implementing **ECDH key exchange and AES-GCM encryption** via the Web Crypto API, deployed on Netlify with persistent encrypted chat history.
- Designed secure cryptographic primitives in TypeScript with support for encrypted file storage and message analytics.

CERTIFICATIONS

Google AI Professional Certificate — Google

Completed 2025

WORK EXPERIENCE

PR & Marketing Head, Chords & Co. — MIT Manipal

July 2025 – Present

- Secured **₹10,000 in sponsorships** for Rhapsody '26, a **20% increase** over the previous tenure, through direct outreach and partner negotiations.
- Recruited, interviewed, and trained a **20+ member PR team**; orchestrated campus-wide publicity campaigns across posters, info desks, open mics, and interactive quizzes driving **800–1,000 attendees** across the 5-day event.